Principles of Machine Learning

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lecture 01

a first glimpse at machine learning . . .

artificial intelligence (AI) is a technical discipline that aims at building *cognitive machines*

$\Leftrightarrow$ AI is concerned with computer algorithms and systems that can see, hear, read, write, make plans, guide robots, . . .

machine learning (ML) is an important branch of artificial intelligence

this lecture provides a first rough overview of the *what*, *why*, and *how* of machine learning

in passing, we already encounter some of the terminology commonly used in the field

lecture 01
and the machine learning pipeline

we also have a first look at the different stages involved in *training* a computational intelligence system to perform cognitive tasks

in short, we look at what people call the machine learning pipeline and emphasize some of its general aspects we must be aware of

outline

machine learning
- *what*, *why*, and *how* ?

the machine learning pipeline
- data pre-processing
- model selection
- model fitting
- model evaluation

summary

what is machine learning ?

machine learning is the science of
fitting mathematical models to data



machine learning is to run computer algorithms
on exemplary data to adjust the parameters of
other computer algorithms ($\Leftrightarrow$ models) so that
these become able to perform cognitive tasks

the above definition deliberately *demystifies* the notion of ML

indeed, dramatic progress over the past decade has caused considerable public interest in ML but accounts in the media are usually horribly ignorant and hand-wavy

however, to make sense of our definition, we need to answer two obvious follow up questions, namely

1) *what* are data ?
2) *what* are mathematical models ?

definitions of the concept of *data* can be surprisingly multi-faceted
and may range from more philosophical to more technical aspects

for instance, we may say that . . .

philosophically, data are known facts that can be recorded, stored, and retrieved and have an implicit “meaning”

technically, data are collections of (numerical) values that result from measurements, surveys, or similar procedures and can be processed by computers

we also distinguish

structured data

- $\Leftrightarrow$ data which easily fit into spreadsheets or relational data bases
- $\Leftrightarrow$ data which naturally allow for organizing, searching, or hashing
- $\Leftrightarrow$ data which can be represented in form of elementary data types

unstructured data

- $\Leftrightarrow$ data such as texts, speech recordings, images, videos, ...
- $\Leftrightarrow$ data whose automatic analysis poses challenges

meta-data / annotations

- $\Leftrightarrow$ data which carry additional information about other data

  tags or labels, time when- and location where a picture was taken, captions, text annotations, ...

in this course, our focus will be on ML algorithms and their underlying mathematical principles rather than on ML system engineering

$\Leftrightarrow$ we will study algorithms that process numerical values or data points in order to automatically determine their "meaning"

(in doing so, we will assume a rather mathematical view on "meaning")

speaking about mathematics . . .

mathematical models are formal descriptions
of certain aspects of the observable world



mathematical models provide logical or structural
characterizations of principles, states, or dynamics
of the observable world and thus allow for making
(informed) predictions or decisions



note

mathematical models are typically but coarse abstractions of very complex phenomena whose minute details may be too complicated for a formal description

All models are wrong, but some are useful.
–George Box

question

OK, but isn't machine learning all about neural networks and such ???

question

OK, but isn't machine learning all about neural networks and such ???

answer

well, neural networks are nothing but very general mathematical models

one more thing . . .

there are three major ML paradigms

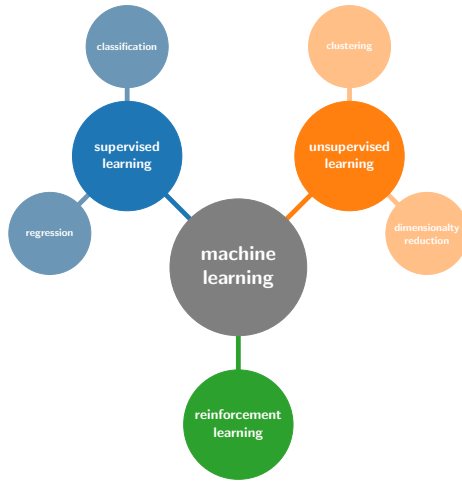
supervised learning

unsupervised learning

reinforcement learning

this course will focus on the first two

for the third, we have another course



why do we use machine learning ?

model fitting allows us to make sense of data or to generalize from known observations to new ones

$\Leftrightarrow$ we typically use machine learning to solve problems such as

- regression
- classification / pattern recognition
- density estimation / data clustering
- $\vdots$
- simplification / surrogate modeling
- (multi-modal) content generation

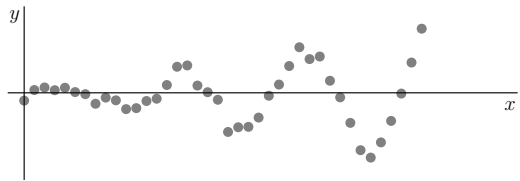
(didactic) examples

regression

given data

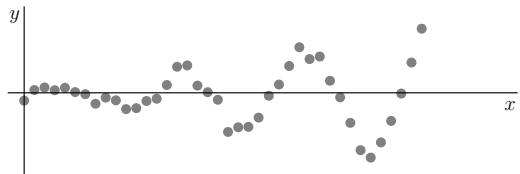
$$D = \left\{ \left(x_j, y_j \right) \right\}_{j=1}^{n}$$

where $x_j, y_j \in \mathbb{R}$, we may have reason to believe that the y_j depend on the x_j



we may *model* this dependency using a function

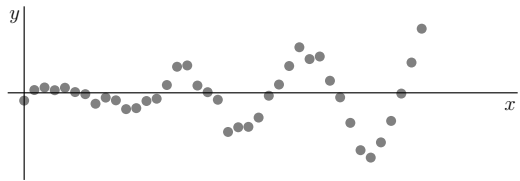
$$f : \mathbb{R} \to \mathbb{R}$$



which (approximately) maps the independent variables or predictors x_j to the dependent variables or outcomes y_j

$$y_j \approx f\left(x_j\right)$$

the problem is that there are infinitely many functions f

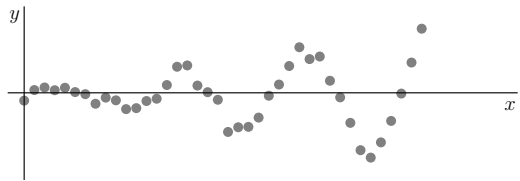


$\Rightarrow$ we “somehow” need to narrow down our choices of f

$\Leftrightarrow$ we “somehow” have to *hypothesize* a *suitable model*

here, we *hypothesize* that this family of functions

$$f\big(x \mid a, b\big) = x^a \cdot \sin\big(b \cdot x\big)$$

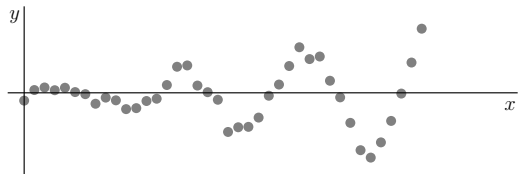


with parameters $a, b \in \mathbb{R}$ may provide an appropriate model of the data in D

$\Rightarrow$ our model fitting task is to determine optimal parameters a and b of $f\big(x \mid a, b\big)$

$\Leftrightarrow$ we need to determine which choice of a and b best matches $f\big(x \mid a, b\big)$ to D

the next problem is that there are infinitely many choices for a and b



$\Leftrightarrow$ our hypothesis class is infinitely large

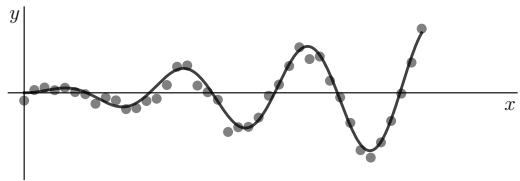
$\Leftrightarrow$ we cannot test every conceivable choice of parameters to determine the optimal ones

$\Rightarrow$ we require smarter model fitting algorithms

indeed, using (simple) statistical optimization techniques, we find that $f\big(x \mid a, b\big)$ with

$$a = 1.10$$
$$b = 1.99$$

characterizes our data very well



principle of machine learning

we may regress anything onto everything

however, when dealing with data such as

$$D = \Big\{ \big(\texttt{"when I came home, my"}, \texttt{"dog"}\big), \\ \big(\texttt{"I came home, my dog"}, \texttt{"was"}\big), \\ \big(\texttt{"came home, my dog was"}, \texttt{"already"}\big), \\ \big(\texttt{"home, my dog was already"}, \texttt{"waiting"}\big), \ldots \Big\}$$

we will obviously have to work with very, very sophisticated models . . .

classification / pattern recognition

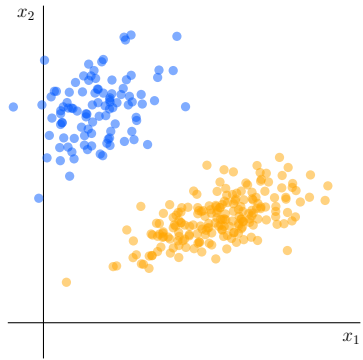
given annotated data

$$D = \left\{ \left(\boldsymbol{x}_j, y_j \right) \right\}_{j=1}^{n}$$

where the data points $\boldsymbol{x}_j \in \mathbb{R}^2$ come from two classes Ω_1 or Ω_2 and the annotations

$$y_j = \begin{cases} +1 & \text{if } \boldsymbol{x}_j \in \Omega_1 \\ -1 & \text{if } \boldsymbol{x}_j \in \Omega_2 \end{cases}$$

are label values that indicate class membership, ...

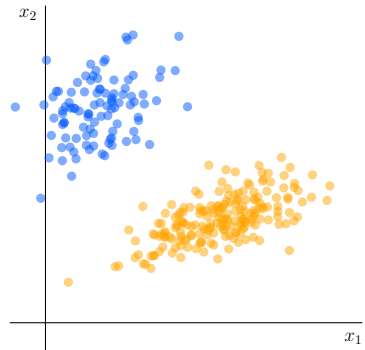


we may want to *train* a classifier

$$f : \mathbb{R}^2 \to \{-1, +1\}$$

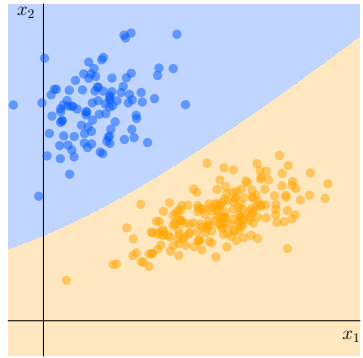
which (approximately) maps the given data points to their labels

$$y_j \approx f(\boldsymbol{x}_j)$$



again, the common approach is to

1) hypothesize a suitable model
2) use statistical optimization to adjust the model parameters such that the model fits the given data

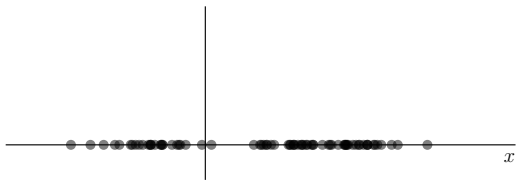


density estimation

given data

$$D = \{x_j\}_{j=1}^{n}$$

where $x_j \in \mathbb{R}$, we might wonder
how these data are distributed



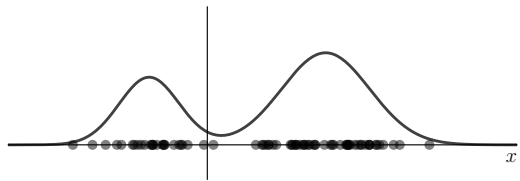
$\Leftrightarrow$ we are interested in a probability density function

$$f : \mathbb{R} \to [0, 1]$$

that characterizes the distribution of the given data

again, we commonly

1) make (reasonable) modeling assumptions
2) apply statistical optimization to estimate parameters such that our model best matches the data



principle of machine learning

it may sound crazy, but nowadays (since ~2015) density estimation is by far the most important machine learning task

this is because all modern generative AI is at its heart based on (variants of) fitted density models

clustering

given data

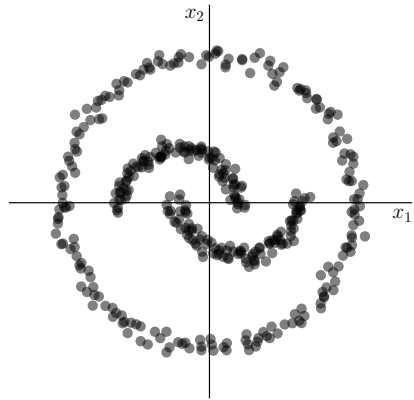
$$D = \{\boldsymbol{x}_j\}_{j=1}^{n}$$

where $\boldsymbol{x}_j \in \mathbb{R}^2$, we might wonder if these data contain any structures, subgroups, or clusters

for instance, if we have reason to suppose there are k clusters, we look for a function

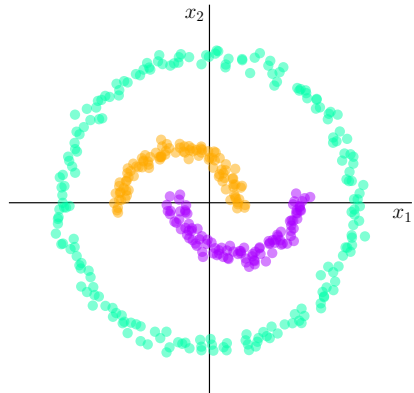
$$f : \mathbb{R}^2 \to \{1, 2, \ldots, k\}$$

which assigns data points to cluster labels



again, the common approach is to

1) hypothesize a suitable model
2) use statistical optimization to adjust the model parameters such that the model fits the data



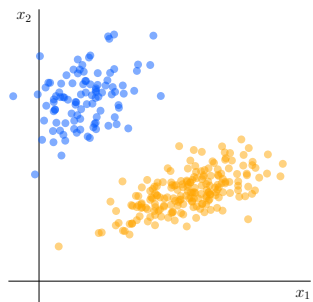


note

regression and classification are common examples of supervised learning problems

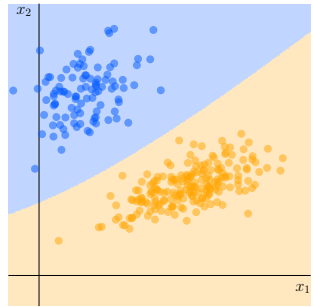
$\Leftrightarrow$ given data consists of pairs $(\boldsymbol{x}_j, y_j)$ of observations $\boldsymbol{x}_j$ and corresponding targets y_j

such pairs explicitly encode (certain) structures within the universe of all possible observations



$\Rightarrow$ the task is to learn an abstract representation of these structures

$\Leftrightarrow$ the task is to learn to replicate and generalize known structures

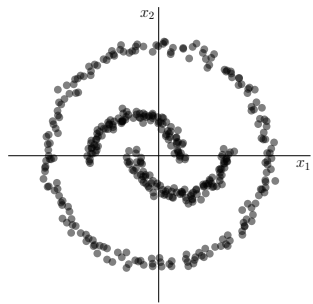


note

density estimation and clustering are unsupervised learning problems

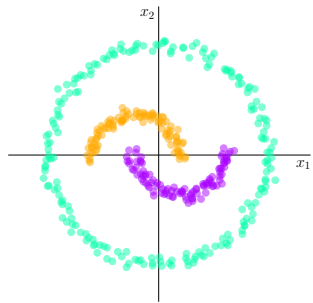
$\Leftrightarrow$ given data consists of observations $\boldsymbol{x}_j$ only

information as to possible structures among these observations is implicit a best



$\Rightarrow$ the task is to identify (any) possible structures

$\Leftrightarrow$ the task is to make implicit information explicit



question

OK, but now for real: *why* do we fit models ?

question

OK, but now for real: *why* do we fit models ?

answer

good models will allow us to automatically

compress information, generalize given examples, reason about cause and effect, make predictions, decisions and plans, generate content ...

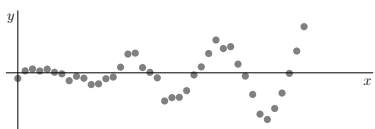
$\Leftrightarrow$ good models will allow us to implement cognitive capabilities on computers

examples

compression / generalization

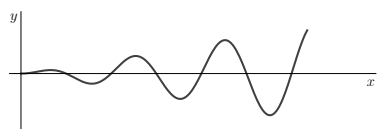
data set (⇔ many numbers)

x_j :	0.00	0.26	0.51	0.77	1.03	1.28	1.54	1.79 $\cdots$
y_j :	−1.47	0.52	1.00	0.49	0.96	0.14	−0.25	−2.06 $\cdots$



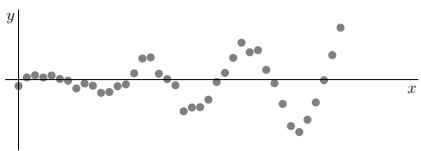
fitted model (⇔ few parameters)

$$y = x^{1.1} \cdot \sin\big(1.99 \cdot x\big)$$

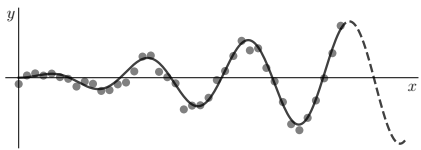


predictions / forecasts

data

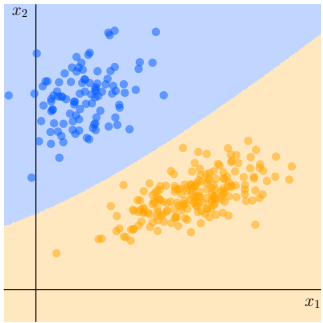


prediction

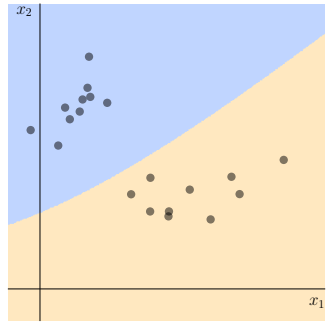


classification / decision making

a trained classifier . . .



can classify novel observations



how do we do machine learning ?

given everything we said so far, there remain three fundamental questions

1) *how* do we chose models ?
2) *how* do we fit them to data ?
3) *what* do we need to be aware of when we do this ?

alas, they are way too far reaching to be answered in an introductory lecture

they are what this course is all about and we shall study them exhaustively

for now, we note ...

principles of machine learning

mathematically, machine learning involves

- geometry and linear algebra
- probability theory and statistics
- calculus and optimization techniques
- ⋮

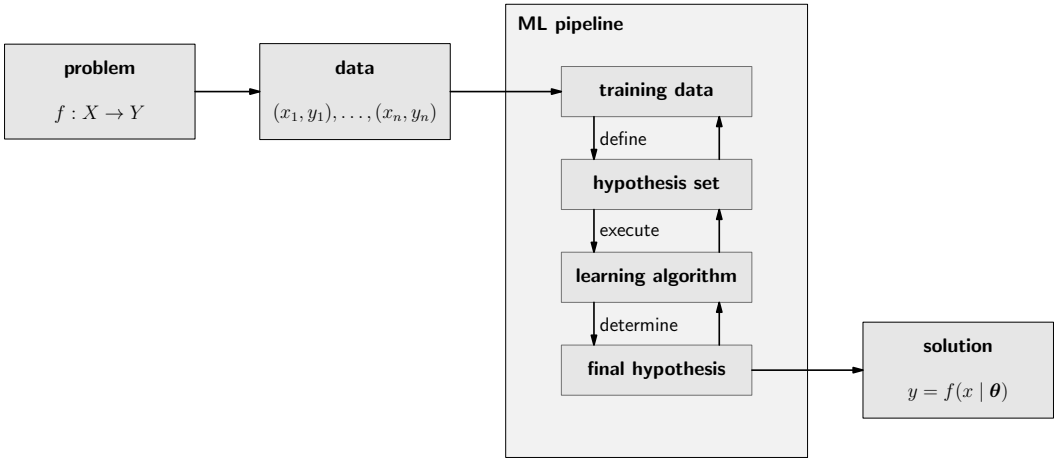
computer scientifically, machine learning involves

- systems for storing, curating, accessing, and processing data
- efficient and robust implementations of model fitting algorithms

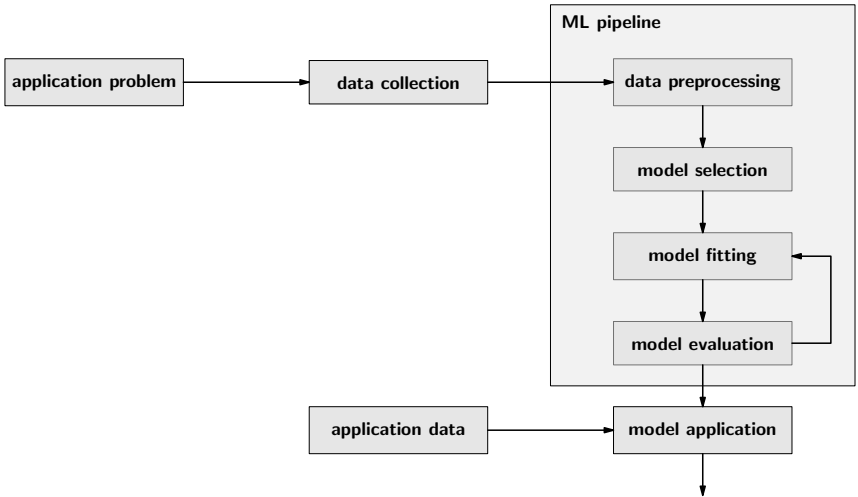
speaking of *system*s . . .

the machine learning pipeline

a machine learning pipeline (von Rueden et al., IEEE Trans. KDE, 2021)



another machine learning pipeline



question

wait a minute, why are there two different pipelines ?

question

wait a minute, why are there two different pipelines ?

answer

there might be even more such pipelines, because . . .



note

to a certain extent, *machine learning is an art rather than a science*

or, stated less defiantly, *machine learning is still a young discipline*

$\Rightarrow$ ways of talking about aspects of machine learning are not as fixed as the nomenclature or terminology of more established sciences

$\Leftrightarrow$ authors (of textbooks, papers, . . .), experts, and practitioners still use different terminology to refer to basically the same ideas . . .

this is unfortunate, but we must live with it

for example

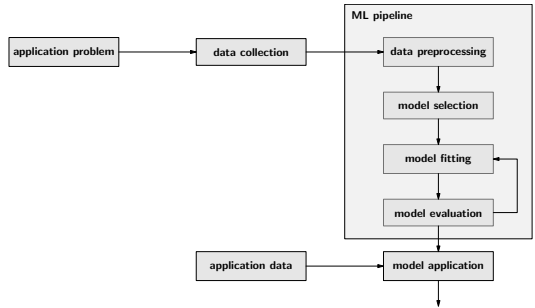
- some talk about the *hypothesis set*, others use the term *model family*
- some talk about the *final hypothesis*, others use the term *fitted model*

principle of machine learning

training a computational intelligence system to automatically perform (difficult) cognitive tasks is a *process*

this process commonly involves several consecutive stages

- problem specification
- data collection
- data pre-progressing
- model selection
- model fitting
- model evaluation
- model application



problem specification

first of all, we need to clarify what application problem
we want to solve by means of machine learning

do we intend to recognize cats and dogs in images ?

do we want to predict the likely yield of a crop based
on nutrient measurements and weather data ?

⋮

problem specification / data collection

first of all, we need to clarify what application problem
we want to solve by means of machine learning

do we intend to recognize cats and dogs in images ?

do we want to predict the likely yield of a crop based
on nutrient measurements and weather data ?

⋮

once the goal is clear, we require corresponding data

which we record ourselves, obtain from collaborators, find on the Web, . . .



note

sometimes, these two initial stages are reversed

$\Leftrightarrow$ we might have data and need machine learning to make sense of it

say, we are given a humongous set of high dimensional data points which overwhelms our human senses and cognitive capabilities ...

we may then want to use machine learning for knowledge discovery

$\Leftrightarrow$ we may want to automatically determine if the data contains patterns, correlations, or cause and effect relations which we should know about

data pre-processing

turning raw data into training data may require pre-processing such as

data cleansing
⇔ removal of outliers, incomplete- or faulty data points, etc.

data labeling
⇔ (manual) categorization or annotation of given observations

feature extraction
⇔ transformation of observations into more abstract representations

normalization
⇔ transformation of representations into “manageable” numerical ranges

data labeling can be easy but is typically cumbersome

> for instance, for some scenarios, it may suffice to store pictures of cats and dogs in folders called “Cats” and “Dogs” but other scenarios might require us to carefully indicate which pixels in an image belong to a cat or to a dog . . .
>
> nowadays, we can of course get assistance from *foundation models* but must trust that they work reliably . . .

annotated data is incredibly valuable

> data which have been (reliably) sorted or categorized or augmented with additional information (meta data) by human experts or AIs can make or break success of machine learning algorithms

feature extraction either requires background knowledge . . .

for instance, observations of body heights and weights of people can be turned into body mass indices which are known predictors for diseases

. . . or can be accomplished automatically

given enough data, deep neural networks can *learn* useful representations

model selection

given a sample of problem specific data, we need to decide for a “suitable” model class

$\Leftrightarrow$ we need to specify a family of mathematical functions which we expect to be able to solve the problem at hand

common models will be studied later on

model fitting / training

having specified a model class, we need to determine or learn "appropriate" model parameters

$\Leftrightarrow$ we need to run optimization algorithms to fit the (mathematical) model to the given data

model training will be studied later on

model evaluation / testing

having fitted a model to the data, we should evaluate if it can generalize

$\Leftrightarrow$ we should test whether the model only explains the training data or also works well on previously unseen test data

principles of machine learning

training data must be *representative*

$\Leftrightarrow$ they should reflect all relevant aspects of what is to be learned

(to see what happens if they aren't, consider the 2015 Google photos incident)

test data must be *independent* from the training data

$\Leftrightarrow$ when working rigorously, we really must insist on $D_{trn} \cap D_{tst} = \varnothing$

in practice, these two principles are ridiculously important, because ...

principles of machine learning

good performance of a model on its training data means nothing

what really counts are generalization / extrapolation capabilities of a trained model

$\Leftrightarrow$ what really counts is how well a trained model performs on independent test data

we must avoid the beginner's mistake of improper model testing

$\Leftrightarrow$ any statement we make about the performance of a machine learning system must be derived from test data which is independent of training data



note

however, things have obviously changed

modern *AI agents* and *genAI models* make the above look old school

nowadays, cutting edge models are trained with trillions of data points

for these, traditional concerns about *over-fitting* are not quite so severe

alas, as of today, only a few players across the world are technically able ($\Leftrightarrow$ have the compute infrastructure) to work in these settings

$\Rightarrow$ **unless you are with OpenAI, DeepMind, . . . the above is still important**

model refinement

if a trained model or system does not perform well (enough),
we may need to

- consider additional / better training data
- reconsider our model class / hypothesis set
- ⋮

$\Leftrightarrow$ we may need to iterate through (parts of) the pipeline again

w.r.t. foundation models, this is also called model *fine-tuning*

model application

if a trained model or system performs and generalizes well, we can apply it in practice

$\Leftrightarrow$ we can feed new data from our application domain into the system and work with the predictions it generates

summary

we now know that

machine learning is the science of fitting models to data

machine learning terminology is often still rather fuzzy

machine learning is not an end in itself but a tool(box)
for problem solving in data analysis and synthesis

a machine learning pipeline involves different stages such as

- data pre-processing, model selection, model fitting, model evaluation

a picture says a 1000 words . . .

